

6.622 Design Project: 15V 150W Buck Converter

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1 Design Overview

The 15V 150W PWM buck converter in Figure 1 operates in continuous conduction mode with a switching frequency of $500kHz$, using the electrical components in table 1, meeting the specifications shown in Table 2

The converter uses a PI to maintain a constant output voltage with the transfer function shown in Equation 1 where $P = 0.02$ and $\omega_L = 2\pi \times 5000 \text{ s}^{-1}$.

$$G_c(s) = P(1 + \frac{\omega_L}{s}) \quad (1)$$

For thermal management the converter has a single 3 inch 60470 aluminum extrusion heat sink mounted to the back of the converter PCB. Heat from the MOSFET and Diode is conducted to the other side of the PCB through a 5x5 array of 12mil thermal vias on the largest pad of each device. A SIL-PAD 400 thermal pad is also used to electrically isolate the sink from the PCB.

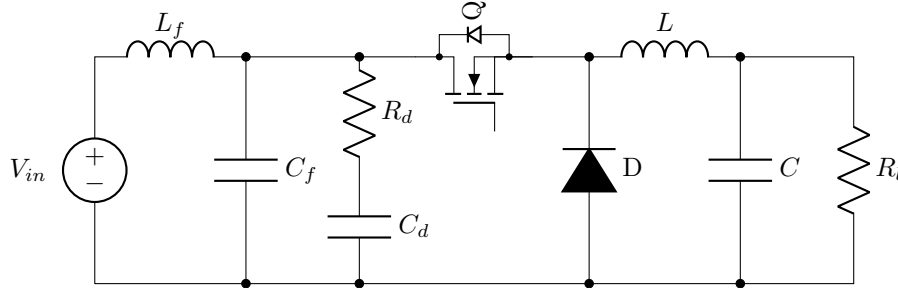


Figure 1: Buck Converter Schematic

Designator	Value	Part Number
L_f	1.6uH	RM8/I-F3F-A400, 2 turns 10AWG
C_f	20uF	206K050CS4
C_d	100uF	UHD1H101MHD
R_d	150mΩ	parasitics from C_d
Q	–	NTMFS5844NL
D	–	PMEG045V100EPD
L	1.6uH	RM8/I-F3F-A400, 2 turns 10AWG
C	22uF	KTS500B226M78N0T00

Table 1: Electrical Components

Parameter	Specified	Design
Output Voltage (static)	$15V \pm 3\%$	$15V \pm 0.1\%$
Output Voltage (transient)	$15V \pm 20\%$	$15V \pm 11.5\%$
Output Voltage Ripple (V_{pp})	80mV	53mV
Input Current Ripple (A_{pp})	50mA	40mA
Minimum Efficiency	90%	95.7%

Table 2: Specifications

2 Active Component Selection

The deciding factor for the selection of passive components is the switching frequency. A higher frequency gives lower losses and lower ripple in the passives, so when possible higher frequencies are desirable. However, a higher switching frequency creates larger losses in the active components. These losses can be split into two categories, conduction losses and switching losses.

For sufficiently fast switching diodes the diode switching losses can be ignored with the conduction losses modeled in Equation 2.

$$P_{d,max} = V_{fd} I_{out,max} (1 - D) \quad (2)$$

This creates relatively simple design criteria for choosing a diode, find one with the lowest forward voltage. However, the diode must also be able to handle an average current of $(1 - D)(10A) \approx 5.3A$ and the 10A peak inductor current. It also must hold off a reverse voltage of the maximum 32V input with some safety factor. Given this, the converter uses the PMEG045V100EPD schottky barrier diode. It has an average current handling of 10A and peak of 14A providing adequate current capabilities, and rated for 45V providing enough voltage blocking. The diode has a worst case forward voltage of 0.49V. This results in a worst case power dissipation of 2.6W or 1.7% of the output power.

For the MOSFET, both conduction and switching losses must be considered. Conduction losses can be modeled as resistive as shown in Equation 3. Switching losses are modeled as a linear rise and fall in voltage and current across the device. This loss is calculated by Equation 4.

$$P_{f,cond} = I_{out,max}^2 R_{DS,ON} D \quad (3)$$

$$P_{f,sw} = \frac{P_{out} V_{in} f_{sw}}{2V_{out}} (t_r + t_f) \quad (4)$$

A somewhat arbitrary limit on the active component power dissipation can be derived from the fact that a majority of the losses in the converter will come from active components. Since the converter is required to be 90% efficient, allowing active components to constitute 5% of the losses will give plenty of headroom for passive losses. Given the known diode losses the MOSFET should have $\approx 3.3\%$ or $\approx 4.9W$ of loss.

Table 3 shows what the losses would be for various devices and switching frequencies. To maintain efficiency headroom, the converter uses the NVMFS5844NL at a switching frequency of 500kHz. This generates a loss of 3.62W or 2.4%.

It's also important to ensure these MOSFETs are rated for the required current and voltage. The NVMFS5844NL, being rated for 60V and 61A is capable in both the on and off state, but at the peak of the transition it must withstand the full voltage and full current at the same time. While the NVMFS5844NL cannot do this continuously, this transition only lasts 35ns, something it is also rated for.

Device	Frequency					
	50kHz	100kHz	250kHz	500kHz	750kHz	1MHz
NVMFS5844NL	2.08W	2.53W	3.88W	6.11W	8.36W	10.6W
NVMFS5844NL	1.10W	1.38W	2.22W	3.62W	5.02W	6.42W
NVMFS5C670NL	0.93W	1.44W	2.98W	5.54W	8.10W	10.7W
NVMFS5C628NL	0.67W	1.18W	2.70W	5.24W	7.78W	10.3W

Table 3: Total MOSFET Losses

3 Passive Component Selection

The passive components in the power stage of the converter determine the output voltage and current ripple. The converter should operate in continuous conduction mode, the inductor current ripple is limited to twice

the minimum output current of the converter, and thus an inductor size larger than $1.125\mu H$ as given by Equation 6.

$$I_{out,min} > \frac{I_{L,pp}}{2} = \frac{V_{out}(1-D)D}{2Lf_{sw}} \quad (5)$$

$$L > \frac{V_{out}(1-D)D}{2f_{sw}I_{out,min}} = \frac{(15V)(0.5)(0.5)}{2(500kHz)(3.33A)} = 1.125\mu H \quad (6)$$

The design of this inductor can be modeled using the magnetic circuit shown in Figure 3. This design uses a gaped RM8 3F3 core with a $180\mu m$ gap and 2 turns of wire. This produces a coil inductance of $1.6\mu H$ (Equation 8) which is sufficient to satisfy the continuous conduction mode requirement.

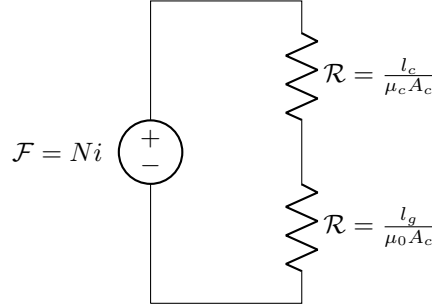


Figure 2: Magnetic Circuit

$$\Phi = \frac{A_c N i}{\frac{l_c}{\mu_c} + \frac{l_g}{\mu_0}} \quad (7)$$

$$L = \frac{N\Phi}{i} = \frac{N^2 A_c}{\frac{l_c}{\mu_c} + \frac{l_g}{\mu_0}} = \frac{(2 \text{ turns})^2 (63.0mm^2)}{\frac{38.4mm}{2000\mu_0} + \frac{180\mu m}{\mu_0}} = 1.6\mu H \quad (8)$$

When experiencing peak current this will generate a maximum B field of $0.19T$ in the inductor core (Equation 9). Luckily this is below the $0.3T$ saturation limit for 3F3.

$$B_{max} = \frac{\Phi}{A_{min}} = \frac{A_c N i_{max}}{A_{min}(\frac{l_c}{\mu_c} + \frac{l_g}{\mu_0})} = \frac{(2 \text{ turns})(63.01mm^2)(13.33A)}{(55.4mm^2)(\frac{38.4mm}{2000\mu_0} + \frac{180\mu m}{\mu_0})} = 0.19T \quad (9)$$

The conductor size used in the coil must be large enough to handle both the average $10A$ DC component of the current through the inductor as well as the $3.33A$ peak AC component. Approximating the DC component as occupying the entire area of the wire and the AC only occupying the one skin depth from the surface a wire with AWG13 or smaller is required to maintain a current density below $500A/cm^2$. This design uses 10 AWG wire which produces a maximum current density of $140A/cm^2$ (Equation 10). Two turns of 10AWG, each with a diameter of $2.6mm$, fit easily in the $8.6mm$ winding window for the RM8 core.

$$J = \frac{I_{DC}}{A_w} + \frac{I_{AC}}{\pi d_w \delta} = \frac{10A}{5.26mm^2} + \frac{3.33A}{\pi(2.6mm)(92.2\mu m)} = 140A/cm^2 \quad (10)$$

This corresponds to a maximum restive power dissipation in the inductor of $53mW$ (Equation 11) and a DC equivalent series resistance of $383\mu\Omega$.

$$\begin{aligned} P_{Diss} &= N l_c \rho_{Cu} \left(\frac{I_{DC}^2}{A_w} + \frac{I_{AC,rms}^2}{\pi d_w \delta} \right) \\ &= 2(42mm)(24.02\mu\Omega m) \left(\frac{(10A)^2}{(5.26mm^2)} + \frac{(3.33A)^2/2}{\pi(2.6mm)(92.2\mu m)} \right) = 53mW \end{aligned} \quad (11)$$

To meet the output voltage ripple requirement the minimum capacitance given by Equation 13 is $10.4\mu F$.

$$V_{out,pp} < V_{c,pp} = \frac{I_{L,pp}}{8Cf_{sw}} \quad (12)$$

$$C > \frac{(3.33A)}{8(80mV)(500kHz)} = 10.4\mu F \quad (13)$$

Adding in some safety factor the design uses the KTS500B226M78N0T00 $22\mu F$ ceramic capacitor which is rated for $50V$ providing a large headroom for the $15V$ output. It also has a large ripple capability of $3A_{rms}$ which provides a modest headroom for the $3.33A_{pp} \approx 2.35A_{rms}$ that the capacitor experiences. Because the design uses a very low ESR ceramic capacitor power dissipation is negligible.

Given the chosen inductance and capacitance the peak to peak output voltage ripple can be calculated to be $V_{out,pp} = 53mV$ (Equation 14)

$$V_{out,pp} = \frac{I_{L,pp}}{8Cf_{sw}} = \frac{V_{out}(1-D)D}{8LCf_{sw}^2} = 53mV \quad (14)$$

4 Input Filter Design

The converter uses a simple LC input filter with a damping resistor and capacitor. The filter is only on the high side, assuming the converter had a very low impedance to ground utilizing a ground plane layer of the PCB, and a small physical distance from the input voltage source.

The MOSFET current that needs to be filtered is a square wave with with peak to peak magnitude of $10V$, corresponding to a fundamental frequency magnitude of $6.366A$ (Equation 15).

$$I_{x,1} = \frac{I_{L,max}}{\pi} \sqrt{2 - 2\cos(2\pi D)} = \frac{2(10A)}{\pi} = 6.366A \quad (15)$$

Since the design is constrained to have a input ripple magnitude of $50mA$, the filter should have a gain of $\frac{0.05mA}{6.366A} = 0.00798 = -42dB$ at the switching frequency of $500kHz$. To provide extra headroom, a gain of $-50dB$ is used corresponding to a filter cutoff frequency of $\omega_f = 1.76 \times 10^5$ via the relationship outlined in Equation 16.

$$\frac{\omega_f^2}{\omega_s^2} = \frac{|I_{in}|}{|I_x|} \quad (16)$$

In addition to the cutoff frequency, it is also desirable to have a filter with low impedance relative to the converter. The effective converter impedance looking from the input port is found in Equation 17 is -1.5Ω . Therefore, the filter should be implemented such that $r_{IN} \gg Z_f = \sqrt{\frac{L_f}{C_f}} = 0.3\Omega$.

$$r_{IN} = \frac{dV_{in}}{dI_{in}} = -\frac{V_{in}^2}{P_o} = -\frac{15^2}{150} = -1.5\Omega \quad (17)$$

From the desired frequency cutoff and impedance the inductor and capacitor values can be determined in equation 18 and 19.

$$C_f = \frac{1}{Z_f\omega_f} = \frac{1}{(0.3\Omega)(1.76 \times 10^5)} = 18.7\mu F \quad (18)$$

$$L_f = \frac{Z_f}{\omega_f} = \frac{0.3\Omega}{1.76 \times 10^5} = 1.7\mu H \quad (19)$$

For the capacitor choice a low ESR is preferred, but the input filter capacitor has to handle larger currents so the 206K050CS4 film capacitor is used. The capacitor is $20\mu F$ and rated for $50V$ giving reasonable headroom to the $32V$ maximum input. Additionally the capacitor is rated for $17.8A_{rms}$ at $500kHz$ giving significant headroom for the $6.366A_{rms}$ it will experience. It also has an input ripple limit of $0.3V_{rms}$ at $500kHz$ which is below the $0.1V_{rms}$ ripple the capacitor will experience as calculated to Equation 20.

$$V_{C,rms} = \frac{1}{C} \int I_{C,rms} dt = \frac{I_{C,rms}}{2\pi(500kHz)(20\mu F)} = 0.1V_{rms} \quad (20)$$

The losses from the equivalent series resistance of the film capacitor are small but not negligible. Assuming the capacitor experiences the AC component of the current drawn through the MOSFET the losses can be modeled by Equation 21 as a dissipation of $62.5mW$.

$$P_{Diss} = I_{AC,rms}^2 \times (5A)^2(2.5m\Omega) = 62.5mW \quad (21)$$

For the filter inductor, since the required inductance is close to the inductor already designed for the power stage, and the filter inductor will experience a low peak current the same inductor design can be used to simplify the converter design overall. This means the inductor will have a the same equivalent DC resistance of $383m\Omega$, but due to the lower current will only dissipate $14mW$.

To damp this filter a series capacitor and resistor are used. The capacitor is chosen to be much larger than the filter capacitor by a factor $\xi = 5$ For a total capacitance of $100\mu F$. Using this factor the optimal damping resistance is given to be $97m\Omega$ by Equation 22.

$$R_d = Z_f \sqrt{\frac{(2+\xi)(4+3\xi)}{2\xi^2(4+\xi)}} = \sqrt{\frac{1.6\mu H}{20\mu F}} \sqrt{\frac{(2+5)(4+3(5))}{2(5)^2(4+5)}} = 0.154\Omega \quad (22)$$

Since a very large capacitance is required, an electrolytic capacitor is ideal. The UHD1H101MHD $100\mu F$ $50V$ electrolytic has impedance between $75m\Omega$ and $220m\Omega$ which is already right in the range of ideal series resistance so a discrete series resistor is unnecessary. Additionally very little current should ever pass through this series resistance so it can be ignored for the purposes of power analysis.

Combined, the filter has a maximum impedance of 0.530Ω given by Equation 23. Since this remains small compared to the converter input impedance it means the filter can be ignored for transient analysis.

$$Z_{max} = Z_f \frac{\sqrt{2(2+\xi)}}{\xi} = \sqrt{\frac{1.6\mu H}{20\mu F}} \frac{\sqrt{2(2+5)}}{5} = 0.530\Omega \quad (23)$$

From this, the actual maximum current ripple is $40mA$.

$$I_{in,pp} = \frac{I_{out,pp}}{\omega_s^2 C_f L_f} = \frac{(12.72A)}{4\pi^2(500kHz)^2(1.6\mu H)(20\mu F)} = 40mA \quad (24)$$

5 Power Dissipation and Efficiency

The power dissipation of individual components in the converter have already been calculated, so the total maximum dissipation can be found by adding them (Equation 25). The total maximum loss is $6.4W$ corresponding to an efficiency of 95.7%

$$P_{total} = P_Q + P_D + P_L + P_{Lf} + P_{Cf} = 3.6W + 2.6W + 53mW + 14mW + 62.5mW = 6.4W \quad (25)$$

6 Thermal Management

There is relatively little power dissipation in the passive components of this converter so they don't produce significant heat. This design uses a single heat sink for the whole converter that is mounted to the bottom of an FR4 PCB. This gives a thermal system modeled by the thermal circuit in Figure 6 for the maximum power dissipation of the active components. The thermal circuit shown uses a SIL-PAD 400 thermal pad between the PCB and a 3 inch length of 60470 aluminum extrusion heat sink.

The constraints of the project dictate a maximum junction temperature rise from ambient of $110^\circ C$ for both the MOSFET and diode. From this a maximum thermal resistance through the PCB can be found for both the MOSFET ($R_{Q,\theta,mb-S} < 21^\circ C/W$) and Diode ($R_{D,\theta,mb-S} < 28^\circ C/W$). Using a 5×5 array of $12mil$ thermal vias, with a typical thermal resistance of $267^\circ C/W$ creates a thermal resistance through

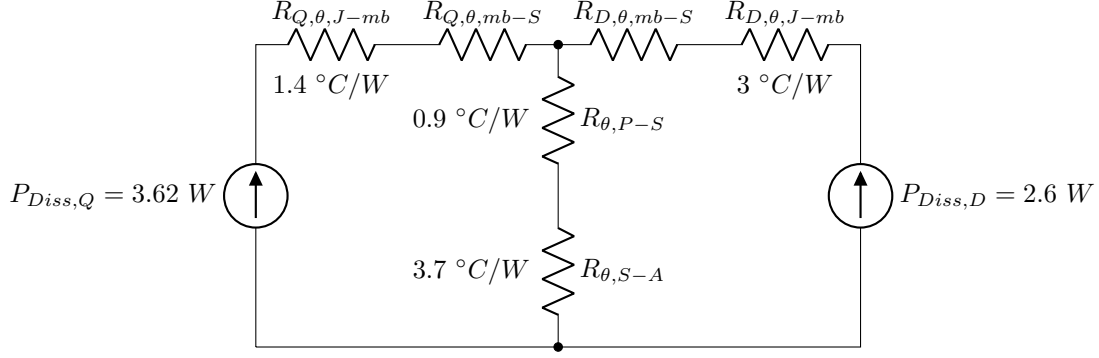


Figure 3: Thermal Circuit

the board of about $10.44\text{ }^{\circ}\text{C/W}$ which is well within the required range for both the MOSFET and Diode. This array would take up approximately $3\text{ mm} \times 3\text{ mm}$ which is small enough to fit in the footprint for both the MOSFET and diode. With this configuration the maximum junction temperature of the MOSFET is $112\text{ }^{\circ}\text{C}$ and the maximum diode junction temperature is $104\text{ }^{\circ}\text{C}$. Both of which are well within the maximum junction temperature limit of $175\text{ }^{\circ}\text{C}$.

7 Open-Loop Transient Response

Because the input filter has been constructed to have a relatively small impedance when compared to the impedance of the converter, it can be ignored for the purposes of determining the state space model of the converter. This model can be represented graphically as a schematic with dependent sources in Figure 7, or the system of differential Equations 26 and 27.

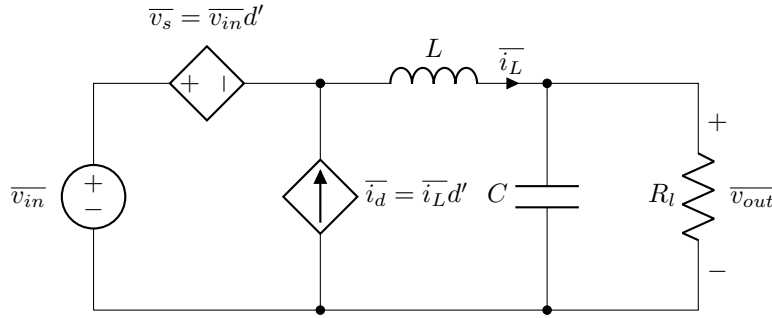


Figure 4: Local Average Schematic

$$\frac{d\overline{i_L}}{dt} = \frac{\overline{V_{in}}d}{L} - \frac{v_o}{L} \quad (26)$$

$$\frac{d\overline{v_o}}{dt} = \frac{\overline{i_L}}{C} - \frac{v_o}{RC} \quad (27)$$

For a constant input voltage and duty cycle, but a change in load this model can be linearized to show the transient response of the converter. Equation 28 shows the transfer function where G is the conductance of the output load.

$$\frac{\tilde{v}_o}{\tilde{g}} = V_o \frac{sL}{s^2LC + sLG + 1} \quad (28)$$

Given the initial conditions that $\tilde{v}_0(0) = 0$ and $\tilde{i}_C(0) = V_0\tilde{g} = C\frac{d\tilde{v}_0}{dt}(0)$ the transient response can be modeled by Equation 29.

$$\tilde{v}_o = V_0\tilde{g}\sqrt{\frac{L}{C}}e^{\frac{-Gt}{2C}}\sin(\frac{1}{\sqrt{LC}}t) \quad (29)$$

The largest magnitude of peak occurs when switching from the largest load of 150W to the smallest load of 50W giving a maximum deviation magnitude of 2.74V. Applying the maximum peak to peak voltage ripple on top of this the maximum voltage deviation during a transient is about 1.72V or 11.5% of the nominal output voltage, a value well under the specified 20% deviation.

Simulating this open loop response in LTSpice, shown in Figures 5 through 8 verify these results for both a step up in power output and a step down through the full range of input voltages.

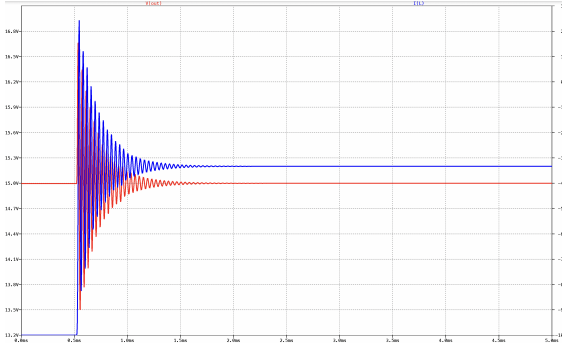


Figure 5: $V_{in} = 24V$, 150W to 50W

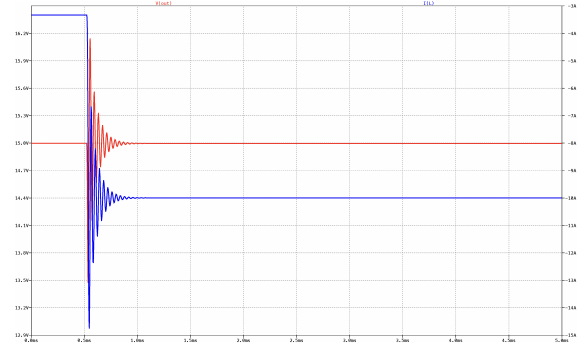


Figure 6: $V_{in} = 24V$, 50W to 150W

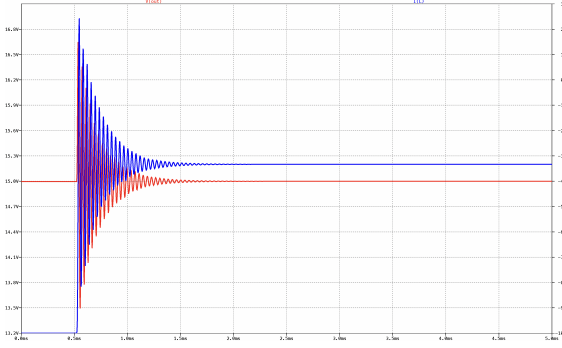


Figure 7: $V_{in} = 32V$, 50W to 150W

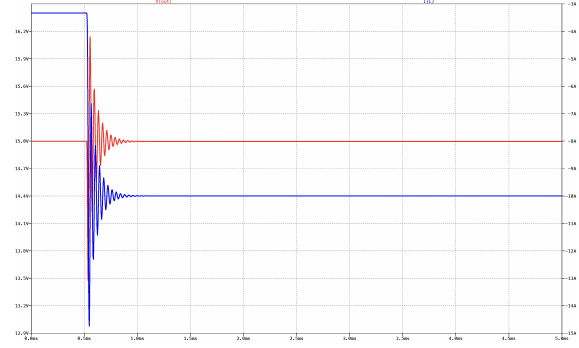


Figure 8: $V_{in} = 32V$, 150W to 50W

8 Closed-Loop Controls

Using the locally averaged model above the system can instead be linearized around a constant load and changing duty cycle for analyzing closed loop controls shown in Equation 30.

$$G_{vd}(s) = \frac{\tilde{v}_o}{\tilde{d}} = \frac{V_{in}/LC}{s^2 + s/RC + 1/LC} \quad (30)$$

For feedback control this circuit uses a PI compensator having the transfer function shown in Equation 31 and the loop gain of equation 32

$$G_c(s) = \frac{\tilde{d}}{\tilde{v}_{err}} = P(1 + \frac{\omega_L}{s}) \quad (31)$$

$$T(s) = G_c(s)G_{vd}(s) = \frac{\tilde{v}_o}{\tilde{v}_{err}} = PV_{in} \frac{1 + \frac{\omega_L}{s}}{s^2 LC + s \frac{L}{R} + 1} \quad (32)$$

At high frequencies the loop gain can be approximated by the top and bottom dominate terms shown in Equation 33

$$T(s) = \frac{\tilde{v}_o}{\tilde{v}_{err}} = \frac{PV_{in}\omega_0^2}{s^2} \quad (33)$$

To avoid switching noise, we can choose a control frequency an order of magnitude below the switching frequency. Assuming a gain of 1 at this frequency the proportionality constant can be solved for in Equation 34.

$$P = \frac{\omega_0}{V_{in}\omega_c} = \frac{1}{\frac{(1.6\mu H)(22\mu F)}{2(2\pi \times 50kHz)^2}} = 0.02 \quad (34)$$

ω_L is chosen to be several orders of magnitude below ω_s in this case $2\pi \times 5000 \text{ s}^{-1}$. Using this controller the resulting transient response is largely unchanged from what was calculated in open loop still remaining within the required range.

Simulating this in LTSpice the transient responses look very close to the open-loop response, though it takes the converter slightly longer to ring down due to the integral component of the feedback. This component, however, ensures the converter will reach it's desired output.

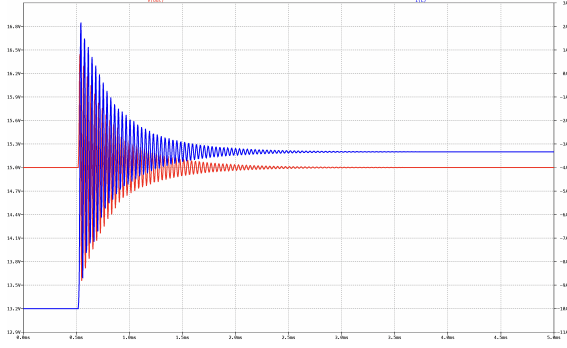


Figure 9: $V_{in} = 24V$, 150W to 50W

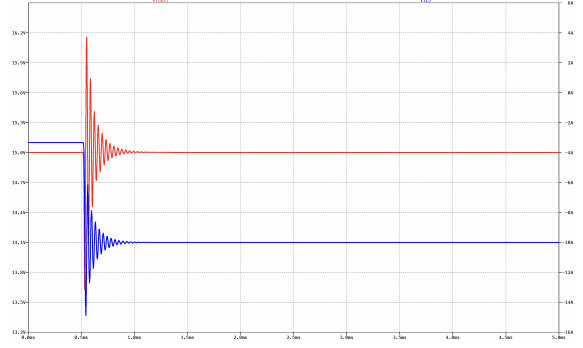


Figure 10: $V_{in} = 24V$, 50W to 150W

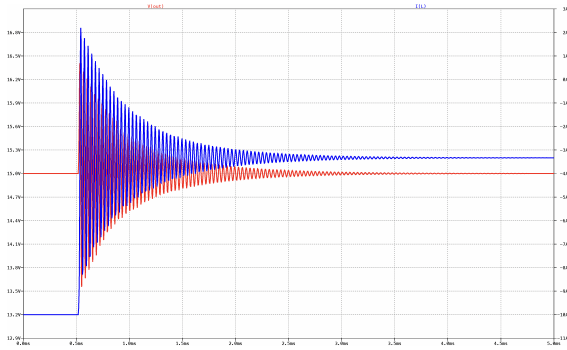


Figure 11: $V_{in} = 32V$, 50W to 150W

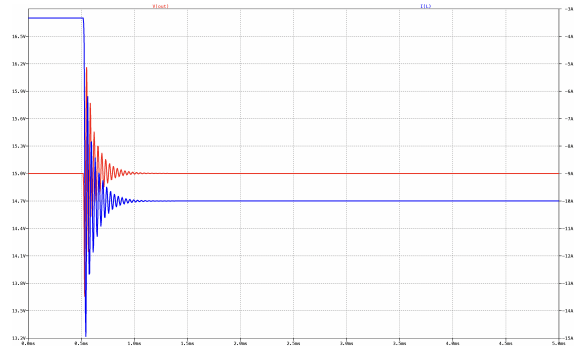


Figure 12: $V_{in} = 32V$, 150W to 50W